CYBERSECURITY & ETHICAL HACKING INTERNSHIP

TASK 1: FOUNDATION & ENVIRONMENT SETUP

Submitted By

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Submission Date

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1. Introduction

Cybersecurity is the practice of protecting systems, networks, and data from cyber threats and unauthorized access. With the increasing dependence on digital technologies, cybersecurity plays a critical role in ensuring the confidentiality, integrity, and availability of information.

The objective of Task 1 is to build a strong foundation in cybersecurity concepts, Linux fundamentals, networking, cryptography, and security tools while setting up a professional cybersecurity lab environment.

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2. Lab Environment Setup

To perform cybersecurity experiments safely, a virtual lab environment was created using VMware Workstation and Kali Linux.

Tools Used

- VMware Workstation
- Kali Linux
- Nmap
- Wireshark
- Burp Suite
- Netcat
- OpenSSL

Lab Configuration

- Installed VMware Workstation.
- Installed Kali Linux virtual machine.
- Verified internet connectivity and network configuration.
- Configured the virtual environment for cybersecurity learning and testing.

Screenshot 1: VMware Workstation Setup

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Screenshot 2: Kali Linux Desktop

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3. Cybersecurity Fundamentals

CIA Triad

Confidentiality

Confidentiality ensures that sensitive information is accessible only to authorized individuals.

Integrity

Integrity ensures that data remains accurate and unaltered during storage and transmission.

Availability

Availability ensures that systems and data are accessible whenever needed by authorized users.

Common Cyber Threats

Malware

Malicious software designed to damage or disrupt systems.

Phishing

A social engineering attack used to steal credentials or sensitive information.

Ransomware

Malware that encrypts data and demands payment for recovery.

DDoS Attack

An attack that overwhelms a target system with traffic, making services unavailable.

SQL Injection

A web application attack that manipulates database queries.

Brute Force Attack

An attack that attempts multiple password combinations to gain unauthorized access.

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4. Linux Fundamentals

Linux is widely used in cybersecurity because of its flexibility, security, and powerful command-line tools.

Commands Practiced

Command	Purpose
pwd	Display current directory
ls	List files and directories
cd	Change directory
mkdir	Create a directory
touch	Create a file
cp	Copy files
mv	Move or rename files
rm	Remove files
chmod	Change permissions
chown	Change ownership
apt	Package management

Screenshot 3: Linux Commands Execution

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5. Networking Basics

Networking enables communication between devices and systems.

OSI Model

The OSI model consists of seven layers:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

TCP/IP Protocol Suite

TCP/IP is the standard communication protocol used on the internet.

TCP

Provides reliable communication between devices.

UDP

Provides faster but connectionless communication.

IP

Responsible for addressing and routing packets.

DNS

The Domain Name System translates domain names into IP addresses.

HTTP and HTTPS

HTTP is used for web communication, while HTTPS provides encrypted communication using SSL/TLS.

NAT

Network Address Translation allows multiple devices to share a single public IP address.

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6. Cryptography Basics

Cryptography protects information from unauthorized access.

Symmetric Encryption

Uses a single key for encryption and decryption.

Examples:

- AES
- DES

Asymmetric Encryption

Uses a public key and private key pair.

Examples:

- RSA
- ECC

Hashing

Hashing converts data into a fixed-length value.

Common algorithms:

- MD5
- SHA-256

SSL/TLS

SSL/TLS secures communication between clients and servers.

Digital Certificates

Digital certificates verify the identity of websites and organizations.

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7. Tool Familiarization

Nmap

Nmap is a network scanning tool used for host discovery and service detection.

Screenshot 4: Nmap Scan

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Wireshark

Wireshark is a packet analysis tool used to monitor and inspect network traffic.

Screenshot 5: Wireshark Capture

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Burp Suite

Burp Suite is a web application security testing tool.

Screenshot 6: Burp Suite Dashboard

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Netcat

Netcat is a networking utility used for debugging and network communication testing.

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8. Learning Outcomes

Through this task, I learned:

- Fundamental cybersecurity concepts.
- Linux command-line operations.
- Networking principles and protocols.
- Cryptography basics.
- Security tool usage.
- Virtual lab setup and configuration.

The practical experience gained from this task provides a strong foundation for advanced cybersecurity topics such as vulnerability assessment, penetration testing, and web application security.

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9. Conclusion

Task 1 successfully introduced the fundamental concepts of cybersecurity and provided hands-on experience with Linux, networking, cryptography, and security tools. The virtual lab environment established during this task will be used for future tasks involving scanning, vulnerability assessment, exploitation, and incident response.

This task strengthened my understanding of cybersecurity fundamentals and prepared me for more advanced practical exercises in the internship.

GitHub Repository

[Paste Your GitHub Repository Link Here]